Chapter 5 Artificial Intelligence in Healthcare

Vinita Gangaram Jansari

Department of Automotive Engineering, Clemson University, Greenville, South Carolina, USA

5.1 History of Artificial Intelligence in Healthcare

Artificial intelligence (AI) is being widely utilized in the healthcare domain, with significant advancements and applications that have led to an evolution of the healthcare domain to improve patient care and medical research. The evolution has happened over a period of time with rapid advancements in the past several decades with the uptake of AI for drug development and medical decision support systems (Asai et al. 2021). AI techniques such as data mining and machine learning (ML) have been utilized to create robust models to perform classification and predictions on healthcare datasets and assist domain experts in the healthcare domain with their decision-making process. However, the healthcare domain has seen a slower utilization of AI techniques due to the lack of trust in the black box ML models. Another challenging aspect has been

the small size and the higher imbalance in the healthcare datasets. One of the milestones in utilizing AI techniques in healthcare came in 2012 because of the advancements in image classification techniques and the integration of AI in patient treatment for informed and precise diagnosis (Anwar et al. 2022). AI has played a significant role in transforming radiology by enhancing diagnostic accuracy (Kaisanesh 2023). AI is expected to catalyze sustainable global and public healthcare, thereby changing various aspects of clinical medicine (Krishnan et al. 2023). From predictive diagnostics to personalized medicine, drug discovery, and treatment plans, the impact of AI techniques in the healthcare domain is profound, with great potential and promise for improving patient outcomes, advancing medical practices, and assisting medical practitioners in their decision-making process.

Some of the key milestones and developments in AI in healthcare are:

- The early beginnings of AI in healthcare started as early as the 1950s till the 1970s with the influence of Alan Turing and the Turing test (Moor 1976), which facilitated the research and development of AI. Early research in AI focused on rule-based and symbolic learning, which laid the groundwork for the later applications of AI in healthcare. MYCIN (Shortliffe 1977), developed in the 1970s, was the first AI system developed for medical use by Stanford University. MYCIN used rule-based reasoning to diagnose bacterial infections and recommend antibiotics.
- The 1980s to 1990s saw growth and development in AI with expansion to expert systems (Waterman 1985). Systems like Internist-I (Miller et al. 1985) and CADUCEUS (Banks 1986) were developed to assist in diagnosing complex diseases using models that utilized human expert-like reasoning, which also helped identify the limitations of using rule-based AI for healthcare. The late 1980s and the early 1990s also saw the resurgence of neural networks, which led to the transition from rule-based models to learning-based models.
- The 2000s brought enhanced computational power with the advent of big data and the utilization of AI in genomics and personalized medicine (LaValle et al. 2010). The increased computational power and stor-

age capacities allowed for the growth of digital health records, medical imaging data, and large-scale genetic data.

Eventually, the modern era of AI in healthcare emerged in the 2010s with the revolutionization of medical image analysis using convolutional neural networks, which significantly improved the accuracy and efficiency of medical image analysis (Ker et al. 2017). AI models started integrating with electronic health records (EHRs) to identify at-risk patients, perform predictive analysis, and provide personalized treatment plans (Mehta and Devarakonda 2018). AI made its way into clinical practice with surgical robots like the Da Vinci Surgical System (DiMaio et al. 2011), which improved precision in minimally invasive surgeries. Various AI-powered diagnostics tools received regulatory approvals, and clinicians started using them in their practice. This era also led to the development of remote monitoring and telemedicine with the use of wearable devices (Farias et al. 2020). Wearable devices such as smartwatches and continuous glucose monitors enabled real-time health monitoring and chronic disease management. More recently, AI was used in COVID-19 response to predict outbreaks, diagnose infections through imagine and develop treatments and vaccines (Khan et al. 2022).

Incorporating AI technologies, such as ML and deep learning, in medical diagnosis has played an instrumental part in transforming the medical diagnosis process in the past years (Tariq 2024). Integrating AI with traditional healthcare techniques can advance the field of precision and personalized medicine with the opportunity to develop customized treatment plans for patients based on their medical history and genetics (Pendy 2023). A recent example portrays the usage of AI techniques to provide data-driven personalized insights and solutions to patients and healthcare providers for diabetes management (Shukur et al. 2023).

5.1.1 Challenges and Limitations

The challenges with the implementation of AI in healthcare are not only internal but external as well, such as the transformation of healthcare professionals and strategic change management (Petersson et al. 2022), along with the ethical challenges of safety, privacy, and threat to patient preferences (Wu and Liu 2023). The AI tools could lead to healthcare in-

equalities and depersonalize the healthcare domain (Babel et al. 2021). Ethical and legal challenges related to data privacy, medical supply chain operations, and health data maintenance continue to be significant hurdles in the application of AI in healthcare (Chikhaoui et al. 2022). Patients are concerned about the adoption of AI in the healthcare domain, leading to slower adoption of AI techniques (Richardson et al. 2021). Also, healthcare professionals have limited exposure to AI, which poses challenges to the effective utilization of AI in the healthcare domain (Shinners et al. 2023).

Geographically, the lack of open medical datasets and cultural biases in algorithms hinder the progress of AI-driven healthcare solutions (Nizam and Aslekar 2021). The underrepresentation of certain populations and medical conditions in the training data for AI applications affects the quality and equity of delivered AI solutions (Parikh and Helmchen 2022). Limitations in the availability of comprehensive biomedical data and legal frameworks restricting collection and access to large-scale medical data pose further challenges to the widespread adoption of AI in the healthcare domain (Pereira et al. 2021). Additionally, it is challenging to integrate AI models with existing EHRs, and developing the IT infrastructure to enable AI systems in the healthcare domain often requires significant investment.

Addressing the challenges related to ethical, legal, and technical hurdles will be crucial in the effective adoption of AI technologies in the healthcare domain to improve patient outcomes and advance AI-driven medical practices (Williamson and Prybutok 2024). The challenges can be addressed through comprehensive guidelines, improved education, and a regulatory framework to harness the full potential of AI in healthcare, ensuring ethical and effective implementation of AI in the healthcare domain.

5.2 New Age of Healthcare with the Use of AI

Cutting-edge technologies of AI are propelling the transformation of healthcare systems toward digital healthcare, enhancing the delivery of healthcare systems (Fosso Wamba and Queiroz 2023). The emergence of

electronic health applications enabled with AI and voice assistants has enhanced healthcare accessibility and convenience in a widespread geographical area (Ahanin et al. 2022). AI has been enhancing nursing practice by providing nurses with the required tools and skills to adapt to the evolving healthcare landscape and provide improved patient care (Rony et al. 2024). With the incorporation of smart wearable sensors, IoT devices, and AI-enabled technologies toward personalized healthcare, more tailored and efficient healthcare services are growing (Junaid et al. 2022).

By leveraging these AI technologies to enhance precision, efficiency, and accessibility, healthcare systems can achieve better outcomes, reduced costs, and improved quality of care. With continued and accelerated innovation, collaboration, and ethical stewardship, AI can usher in a new era of healthcare that is patient-centric, data-driven, sustainable, and accessible.

5.2.1 Large Language Model for Healthcare Domain

Large language models (LLMs) are advanced natural language processing (NLP) models that use deep learning techniques, specifically transformer architecture, with the goal of generating and understanding human-like text. LLMs are increasingly being utilized in the healthcare domain for a variety of applications, showcasing their potential to transform medical practices and improve patient care. LLMs are making a significant impact in healthcare, specifically for patient care, research, and education. LLMs are also being utilized to match patients to clinical trials, improving a critical aspect of healthcare research and treatment (Nievas et al. 2024).

Some of the examples of how LLMs are being utilized in the healthcare domain are:

1. Dental Medicine: LLMs are being utilized in dental medicine for clinical decision support, text summarization, efficient writing, and multilingual communication (Eggmann et al. 2023; Huang et al. 2023a; Rahad et al. 2024).
2. Patient Triage and Clinical Notes: LLMs are being utilized in healthcare for tasks such as assisting in patient triage, drafting clinical notes, completing reimbursement forms, aiding in differential diagnoses, de-

veloping treatment plans, educating patients, and compiling and analyzing research (Jackson and Pinto 2024; Frosolini et al. 2024).

3. Epilepsy Care: LLMs are employed in epilepsy care to extract and analyze patient data, provide clinical summaries, draft patient letters, and offer advice on differential diagnosis and treatment options, enhancing the quality of care in epilepsy care and management (van Diessen et al. 2024; Landais et al. 2024).
4. Ophthalmic Care: LLMs are being evaluated for potential deployment in ophthalmic care, emphasizing the importance of real-world evaluation to determine their effectiveness in clinical trials and practical applications (Biswas et al. 2024; Yaghy et al. 2024; Chen et al. 2024).
5. Vascular and Interventional Radiology: LLMs designed for NLP exhibit promising capabilities in clinical decision-making, workflow optimization, education, and patient-centered care in the field of vascular and interventional radiology (Togunwa et al. 2024; Zaki et al. 2024).
6. Behavioral Healthcare: LLMs have the potential to revolutionize behavioral healthcare by improving engagement and retention through personalized responses, key concept extraction, and addressing patients' unique context and concerns during interventions (Stade et al. 2024; Englhardt et al. 2023; Chiu et al. 2024).
7. Orthopedics: LLMs are utilized in orthopedics for various tasks, including writing, which can significantly enhance the effectiveness and efficiency of surgeons, scientists, and editors in the field (Chatterjee et al. 2023; Brameier et al. 2023).
8. Radiation Oncology: LLM chatbots have the potential to optimize medical practice, reduce workload, and enhance performance in radiation oncology by improving efficiency and streamlining processes (Yalamanchili et al. 2024; Holmes et al. 2023).
9. Medical Education, Research, and Assessment: LLMs are accessible and time-efficient tools that provide vast clinical knowledge, reduce the emphasis on factual recall, and enhance medical education and assessment processes (Yiu and Lam 2023). LLMs like ChatGPT are being used in healthcare education and research to enhance scientific writing, research equity, workflow efficiency, save time in experimental design, aid in drug discovery, and support personalized learning (Sallam 2023).

10. Clinical Documentation and Patient Interaction: LLMs play a role in healthcare settings by assisting in clinical documentation, insurance pre-authorization, summarizing research papers, and answering patient queries through chatbots (Meskó and Topol 2023; Sayin et al. 2024; Goyal et al. 2024; Dada et al. 2024).
11. Assisting in Clinical Decision-making: LLMs are employed to help in patient triage, drafting clinical notes, completing reimbursement forms, aiding in differential diagnoses, developing treatment plans, offering medical recommendations, and improving patient education (Jackson and Pinto 2024; Galitsky 2024).
12. Improving Patient Safety and Pharmacovigilance: LLMs are evaluated for their impact on patient safety and pharmacovigilance, particularly in causality assessment for safety signal detection, adverse reaction management, and regulatory submissions (von Csefalvay 2024; Wang et al. 2023a, 2023b).
13. Specialized Medical Fields: LLMs are utilized in various medical specialties such as ophthalmic care, vascular and interventional radiology, orthopedics, and rheumatology, showcasing their potential in specialized healthcare practices (Biswas et al. 2024; Togunwa et al. 2024; Chatterjee et al. 2023; Yiu and Lam 2023; Venerito et al. 2024).
14. Drug Discovery and Clinical Text Mining: LLMs are leveraged for generating new leads in drug discovery, providing logical feedback, and supporting clinical text mining for structured information extraction from unstructured healthcare texts (Brahmavar et al. 2024; Tang et al. 2023b).

Even with the aforementioned examples of LLMs in healthcare, the lack of domain knowledge has been the biggest challenge in adopting LLMs for specific tasks in healthcare. The lack of domain-specific corpora further hinders the uptake and usage of LLMs for biomedical or clinical data (Yang et al. 2023). To overcome these challenges, a few domain-specific, state-of-the-art, and specialized LLMs have been developed, namely:

1. Bidirectional encoder representations from transformers (BERT) for biomedical text mining (BioBERT) (Lee et al. 2020), as the name suggests, is a specialized variant of the BERT model, which is specifically tailored for biomedical texts and tasks within the biomedical domain.

BioBERT has been pretrained on large-scale biomedical text corpora, which allows BioBERT to capture biomedical domain-specific knowledge and vocabulary. It is open source and freely available, enabling its widespread adoption for various research projects in the healthcare and biomedical domains.

2. Semantic scholar corpora with BERT (SciBERT) (Beltagy et al. 2019) is also a variant of BERT, which has been pretrained on a large corpus of scientific textual data, including research articles from various disciplines such as biology, chemistry, physics, and computer science. SciBERT is also open source and freely available and is being utilized by researchers and the scientific community to advance NLP capabilities in scientific domains like healthcare. SciBERT has been utilized for a wide range of applications in the healthcare domain, ranging from biomedical text mining to clinical decision support systems (CDSS), drug discovery, and patient-centered care.
3. PubMedBERT (Gu et al. 2021) is another variant of BERT, pretrained on a large corpus of PubMed articles, making it suitable for tasks within the healthcare domain. It has been utilized for a wide range of applications, including biomedical text mining, clinical NLP, CDSS, healthcare knowledge graph construction, and drug discovery.
4. ClinicalBERT (Alsentzer et al. 2019) is another variant of BERT developed by researchers at the University of Pennsylvania, which is pretrained on large-scale clinical text data, including EHRs, clinical notes, discharge summaries, and medical literature. It has been used as a tool for processing, analyzing, and understanding clinical text data to enable a wide range of applications in clinical NLP, such as phenotyping, patient stratification, and knowledge graph construction.

These mentioned examples highlight the diverse applications of LLMs in healthcare, demonstrating their potential to enhance patient care and safety, improve medical education and research, and optimize healthcare outcomes by leveraging advanced technologies. By leveraging the capabilities of LLMs, healthcare professionals can improve diagnostics, treatment planning, and overall healthcare delivery, ultimately leading to better patient outcomes and advancements in medical practices.

5.2.2 Generative AI for Healthcare Domain

Generative AI refers to AI systems with the ability to generate new, original content or data that resembles human-like content. Generative AI is increasingly being applied in the healthcare sector, offering innovative solutions and transforming various aspects of medical practices. Following are some examples that showcase the utilization of generative AI methods in the healthcare domain:

1. Radiology: Generative AI methods excel at automatically recognizing complex patterns in imaging data, providing quantitative assessments of radiographic characteristics, and aiding in the interpretation of radiological images, thereby enhancing diagnostic accuracy in radiology (Hosny et al. 2018; Huang et al. 2023b).
2. Digital Medicine: Generative AI powers digital medicine by automating repetitive tasks, allowing healthcare professionals to focus on human-to-human interactions, emotional intelligence, and judgment, thereby improving patient care and the overall healthcare experience (Fogel and Kvedar 2018; Rodriguez et al. 2024).
3. Medical Images: Generative AI applications for medical images are playing an increasing role in biomedical research and healthcare services, aiding in image analysis, diagnostics, and treatment planning (Castiglioni et al. 2021; Pinaya et al. 2023).
4. Clinical Decision-making: Generative AI assists in clinical decision-making by providing insights, aiding in patient triage, drafting clinical notes, developing treatment plans, and improving patient education, thereby enhancing the quality of care provided to patients (Kumar et al. 2021; Rodriguez et al. 2024).
5. Drug Discovery: Generative AI is utilized in drug discovery to generate new leads, optimize drug design, and support pharmacological research, offering innovative solutions in pharmaceutical development and personalized medicine (Thesmar et al. 2019; Bordukova et al. 2024; Tang et al. 2021).
6. Symptom Assessment Applications: Generative AI-based healthcare tools, such as symptom assessment applications, can reduce the burden on healthcare systems, improve patient outcomes, and enhance

the efficiency of healthcare delivery (Millen et al. 2022; Barak-Corren et al. 2024).

7. Automation in Gynecology and Obstetrics: Generative AI techniques are being employed to optimize medical image reconstruction, noise reduction, quality assurance, triage, segmentation, and computer-aided detection in the fields of gynecology and obstetrics, improving diagnostic accuracy and patient care (Weichert et al. 2021; Venkatasubramanian 2022).
8. Precision Medicine: Generative AI in personalized medicine enables precise and effective healthcare interventions, leading to improved patient outcomes, reduced healthcare costs, and enhanced efficiency in healthcare delivery (Ahmed and Adil 2023; Grupac et al. 2023).
9. Smart Healthcare: Generative AI technologies are applied in smart healthcare systems to optimize ICU surveillance, predict mortality, manage hospital resources efficiently, inspect drug reactions, and make critical healthcare decisions during the COVID-19 pandemic (Zaman et al. 2023; Nova 2023).

Generative AI is revolutionizing the healthcare domain by enhancing diagnostic capabilities, improving patient care, optimizing treatment planning, and driving innovation in medical research and healthcare delivery. As can be seen from the aforementioned examples, the new age of AI represents a paradigm shift in how technology has been developed, deployed, and utilized to address societal challenges, drive economic growth, and improve human well-being.

5.3 AI-enabled Medical Devices

AI-enabled medical devices are transforming healthcare by integrating advanced technologies to enhance patient care, improve outcomes, and optimize healthcare services. These devices utilize AI, ML, and data analytics to offer personalized, efficient, and effective healthcare solutions. AI embedded in smart devices is democratizing healthcare by bringing AI-enabled health services, such as CDSS, into the homes of patients (Esmaeilzadeh 2020). Furthermore, AI-based medical devices, especially for image analysis, are becoming increasingly prevalent in medicine, presenting new challenges and opportunities for the application of AI in

healthcare (Pesapane et al. 2018). Medical devices are becoming more effective as they are becoming dynamic and responsive to individual patient needs, a capability that has been facilitated by the integration of AI with medical devices (Arora 2020). AI-based medical devices are also playing a critical role in enhancing the capabilities of the Internet of Medical Things and point-of-care devices used in advanced healthcare sectors such as cardiac measurement, cancer diagnosis, and diabetes management (Manickam et al. 2022). Moreover, AI-enabled medical applications hold the potential to improve diagnosis, treatment, disease prediction, clinical workflow, direct-to-consumer services, genome interpretation, biomarker discovery, and automated robotic surgery (Nolan 2023).

Following are some examples of AI-enabled medical devices:

- Zebra Medical Vision has been utilized to detect various medical conditions, including cancer, liver disease, and cardiovascular issues, with high accuracy with the use of AI techniques (Nadu 2018).
- Aidoc has been used to assist radiologists by detecting abnormalities in medical images and prioritizing critical cases to expedite diagnosis (Stephens 2022).
- IDx-DR has been used to identify diabetic retinopathy in retinal images and has been approved by the FDA for autonomous use (Savoy 2020).
- Qure.ai can interpret head CT scans and chest X-rays with the goal of identifying abnormalities such as fractures, tuberculosis, and intracranial hemorrhages (Engle et al. 2020).
- Tools such as Paige, PathAI, Aiforia, and Proscia assist pathologists in diagnosing diseases accurately and efficiently in oncology (Nakagawa et al. 2023).
- SkinVision provides an AI-based mobile app that can be used to analyze skin lesions and moles to identify signs of skin cancer, along with providing risk assessments and recommendations for further action (Deeks et al. 2020).
- Eko AI, with the integration of digital stethoscopes, analyzes heart sounds to identify murmurs and other cardiac abnormalities (Adedinsewo et al. 2024).

- Proprio Vision integrates AI and augmented reality with the goal of providing surgeons with real-time three-dimensional visualizations of patient anatomy during surgery to improve precision and, thereby, outcomes (Liu et al. 2024).
- Caption Health provides an ultrasound system that helps clinicians interpret cardiac ultrasound images (Driver et al. 2024).
- Tempus utilizes AI to analyze molecular data to aid oncologists in identifying treatment decisions (Osipov et al. 2024).
- IBM Watson for Oncology has provided evidence-based treatment options and recommendations for cancer patients (Somashekhar et al. 2018).
- Google DeepMind's AI for Optical Coherence Tomography (OCT) Scans has been utilized to analyze OCT scans to detect eye diseases such as macular degeneration (Varma 2018).
- SOPHiA Genetics has been utilized to analyze genomic data to identify mutations and biomarkers associated with diseases and provide precision medicine (Guarischi Sousa et al. 2023).
- Viz.ai has been used to analyze brain scans for signs of stroke and to alert specialists, enabling them to perform quicker interventions in emergency care (Karamchandani et al. 2023).
- DreaMed Diabetes has been used to provide personalized insulin dose adjustments based on continuous glucose monitoring of the patient (Nimri et al. 2020).
- Current Health is a remote platform that analyzes data from wearable devices to provide early intervention for patients (Current Health Care-at-home Platform n.d.).

The aforementioned advancements in AI-enabled medical devices highlight the transformative impact of AI on healthcare delivery, patient outcomes, and the quality of care provided to individuals.

5.4 Explainable AI for Healthcare

Explainable AI (XAI) refers to a set of techniques and methods used to make AI systems transparent, understandable, and interpretable to humans. The primary goal of XAI is to provide insights into the decisions,

predictions, and recommendations of AI models to build the trust of users in these models.

Essentially, XAI demystifies the “black box” nature of AI models. XAI plays a crucial role in the healthcare sector, particularly in the development of AI-enabled medical devices. XAI models can be intrinsic or post hoc based on how they operate. Intrinsic models are structured to be understandable, whereas post hoc models provide an explanation for the trained model by analyzing the model along with an external method.

XAI has been applied in various healthcare domains, including medical imaging and healthcare tasks, where deep learning techniques are widely utilized (Allen 2023). XAI techniques enable the interpretation of medical image classification and prediction by highlighting the regions of interest and providing explanations for the model’s diagnostic decisions, enhancing domain users’ confidence and trust in these models. XAI can also identify potential biases or errors in neural networks by providing explanations of the decisions. These applications have the potential to aid in diagnosis, personalized medicine, and clinical decision support, thereby improving the quality of care provided to patients (Sarp et al. 2021). Furthermore, XAI has played a vital role in chronic wound classification, early detection, diagnosis, treatment, outcome prediction, prognosis, and evaluation in healthcare settings (Mansour et al. 2021). Some of the recent XAI models for healthcare data are Grad-CAM, LIME, and SHAP (Panwar et al. 2020; Magesh et al. 2020; Lewin-Epstein et al. 2021).

In the context of disease diagnosis, XAI, in conjunction with the internet of things (IoT) and cloud computing, has transformed conventional healthcare systems into smart healthcare, enabling more efficient and accurate disease diagnosis models (Mohammed et al. 2022). Additionally, XAI has been instrumental in automating digital health systems, thereby enhancing the quality of care services delivered to patients. Moreover, XAI has been utilized in federated learning-based AI approaches in smart healthcare, addressing concepts, taxonomies, challenges, and open issues in the field (Chaddad et al. 2023). By leveraging XAI technologies during the diagnosis phase, clinicians can benefit from improved decision-making processes and diagnostic accuracy (Chaddad et al. 2023).

For drug discovery and development, XAI techniques can analyze genomic and clinical data to identify biomarkers and treatment responses and provide explanations for precision treatment based on individual patient characteristics (Ponzoni et al. 2023). XAI techniques can also provide explanations for predicted drug efficacy, toxicity, and side effects, the guiding development of safer, and more efficient and effective therapies (Kirboğa et al. 2023).

Furthermore, XAI has the ability to analyze EHRs to identify patterns in patient data along with providing explanations for recommended treatment pathways, thereby helping clinicians in making informed decisions about patient care. Particularly, AI is used for CDSS in clinical practice to aid healthcare professionals in disease diagnosis and treatment decisions (Amann et al. 2020). The adoption of AI and XAI-based healthcare devices is on the rise, with over 300 FDA-approved AI/ML-based devices indicating a growing trend in the development of AI-enabled healthcare technologies (Pasricha 2022).

Ensuring data privacy, validation, diverse clinical trials, training resources for patients and healthcare providers, and regulatory oversight are key ethical considerations for trustworthy AI-enabled medical care, and XAI techniques help in achieving these ethical considerations (Mooghali et al. 2023). XAI plays a crucial role in ensuring that AI models are transparent, trustworthy, accountable, and align well with human values and preferences. XAI empowers users to understand, evaluate, and effectively and efficiently utilize AI systems in various domains, specifically healthcare.

5.5 Medical Decision Support Systems

AI-based medical decision support systems utilize AI, ML, and data analytics to enhance clinical decision-making, improve patient outcomes, and optimize healthcare services. These systems aid healthcare professionals in diagnosing diseases, recommending treatment plans, and providing personalized care. AI techniques are increasingly utilized in medical decision support systems to aid physicians in accurate diagnosis, minimize errors, enhance patient safety, and reduce costs (Aljaaf et al. 2014). Medical

decision support systems are incorporating AI and ML approaches to automate disease diagnoses and improve efficiency and accuracy in healthcare settings (Nourani et al. 2020). Furthermore, CDSS, powered by AI, are utilizing patient data to generate treatment recommendations, supporting healthcare providers in making informed clinical decisions (Sharma et al. 2022). AI-driven systems like DESIREE are utilizing clinical data along with AI algorithms for cancer prediction and interpretation, aiding in disease prognosis and treatment planning (Almansour 2022). AI-enabled systems are supporting clinical decision-making for cervical cancer by predicting diagnostic results based on training data, leading to improved accuracy and efficiency in cervical cancer screening (Holmström et al. 2021; Vargas-Cardona et al. 2024).

Moreover, AI-based medical decision-making tools, incorporating fuzzy and soft set theories, are effectively predicting cancer survival and raising disease awareness (Hung et al. 2020). AI technologies are also being applied in ophthalmic diagnosis for clinical support systems, enhancing medical imaging analysis and improving diagnostic accuracy (Han 2022). Recently, the FDA has approved AI devices for disease screening decisions, such as diabetic retinopathy, without the need for clinician interpretation, showcasing the advancement of AI in healthcare (Nsoesie 2018). AI technologies are also improving the efficiency and accuracy of medical diagnosis, clinical data analysis, medical image recognition, and treatment plan decision-making, enhancing personalized care (Xu et al. 2022). Specifically, deep learning algorithms in AI are being utilized for medical image analysis, aiding in the diagnosis and treatment of various diseases, including liver fibrosis and gastrointestinal cancer (Wei and Yang 2021).

AI-enabled medical decision support systems enhance clinical decision-making by providing clinicians with timely, accurate, and actionable insights derived from patient data. By augmenting human expertise with AI-driven analysis and predictions, medical decision support systems have the ability to improve diagnostic accuracy, treatment effectiveness, and patient outcomes.

5.6 Precision/Personalized Medicine Using AI

Precision medicine, also known as personalized medicine, tailors medical treatment and interventions to individual patients based on their unique genetic data, clinical characteristics, and environmental factors. AI has significantly advanced personalized medicine by analyzing large and complex datasets, identifying patterns, and predicting personalized treatment responses (Desai et al. 2024). Some of the examples of how AI is utilized in precision/personalized medicine are:

- Therapy Planning and Risk Prediction: AI techniques are positively impacting personalized care by aiding in therapy planning using clinical, genomic, or social and behavioral determinants of health, as well as risk prediction and diagnosis using genomic or other variables (Chintala 2023). AI models consider patient-specific factors, such as genetic variations, comorbidities, and treatment tolerability, to optimize therapy plans tailored toward individual patient needs.
- Contextualizing Precision Medicine with AI: AI algorithms are performing multimodal data fusion by integrating genomic data, clinical records, medical scans, lifestyle factors, and environmental exposures to generate a comprehensive profile of individual patients (Holzinger et al. 2022).
- Innovative Applications in Precision Medicine: AI is being used in precision medicine for predicting tumor growth, overall survival, identifying biomarkers using biomedical images, and selecting patient populations for clinical practice and clinical trials (Naik et al. 2024).
- Precision Medicine in Neurodevelopmental Disorders: AI is applied in precision medicine for neurodevelopmental disorders to optimize diagnosis, therapeutic intervention, and prognosis using multidimensional biological datasets (Uddin et al. 2019). Genetic testing is being utilized to provide families with personalized genetic risk assessment, helping them understand the genetic basis of neurodevelopment disorders and recurrence risks (Blesson and Cohen 2020).
- Radiogenomics and Precision Medicine: AI-based radiomic and radiogenomic tools are utilized for stratifying patients into precise diagnostic and therapeutic pathways, enhancing personalized treatment in oncology care (Rudie et al. 2019). The integration of radiogenomics

and precision medicine offers insights into disease biology, prognosis, treatment response, and personalized treatment strategies (Shui et al. 2021). By integrating radiomic and genomic data, predictive models can guide treatment decisions, stratify patients into risk categories, and identify candidates for targeted therapies or clinical trials (Bera et al. 2022).

- Cardio-oncology and Precision Medicine: AI and personalized medicine tools present novel pathways in cardio-oncology, offering innovative solutions for individualized patient care (Sadler et al. 2022). Precision medicine integrates circulating biomarkers, such as natriuretic peptides, and inflammatory markers with genomic data to predict cardiovascular risk in cancer patients (Sweeney et al. 2021).
- Future Medical AI Application Requirements: AI is perceived as a transformative technology that will change medicine toward precision and personalized care, revolutionizing healthcare and the role of physicians (Maassen et al. 2021). Future medical AI applications need to include feedback mechanisms, clinician inputs, and user-centered design principles to iteratively refine algorithms and enhance usability.
- AI in Precision Medicine for Health Equity: AI plays a key role in precision medicine by analyzing large datasets, including genetic information, to identify patterns and develop personalized treatment plans for patients (Pendy 2023). AI-driven decision support systems will need to be designed to accommodate cultural, linguistic, and socio-economic diversity by providing personalized healthcare recommendations and educational resources in patients' preferred languages and cultural contexts, thereby improving health outcomes and reducing disparities in healthcare access and utilization (Marjadi et al. 2023).

As showcased by the aforementioned examples, precision/personalized medicine empowered with AI represents a paradigm shift in healthcare, moving toward individualized, proactive, and data-driven approaches for diagnosis, treatment, and disease management.

5.7 Smart Healthcare

Smart healthcare refers to the integration of advanced technologies, data analytics, and innovative solutions to enhance healthcare services, improve patient outcomes, and optimize healthcare delivery. This concept leverages cutting-edge technologies such as AI, IoT, wearable devices, and robotics to transform healthcare delivery, personalize patient care, and promote proactive and preventive healthcare strategies. Smart healthcare aims to improve efficiency, accessibility, and quality of care by utilizing technology to enable remote monitoring, personalized treatment plans, predictive analytics, and data-driven decision-making in healthcare settings. Smart healthcare is rapidly evolving with the integration of advanced technologies and innovative solutions to enhance patient care, improve outcomes, and optimize healthcare services. Here are some of the examples that highlight advancements in smart healthcare:

- 5G-based Smart Healthcare Network: The integration of 5G and IoT is expected to drive next-generation smart healthcare, offering improved connectivity and innovative healthcare solutions (Ahad et al. 2019).
- Smart Nano-inspired Drug Delivery: Rapid advancements in smart drug delivery systems are providing promising avenues to enhance healthcare systems, improve patient outcomes, and enable personalized medicine (Boppana et al. 2024).
- Smart Healthcare in the Age of AI: The combination of smart sensors, actuators, and data analytics in an IoT environment is revolutionizing healthcare, offering real-time monitoring and continuous healthcare services (Nasr et al. 2021).
- Sensors for Context-aware Smart Healthcare: Context-aware smart healthcare systems contribute to enhancing medical services, improving patient welfare, reducing waiting times, and decreasing healthcare costs (Batista et al. 2021).
- Enchanted Life Space in Smart Health: Developments in advanced technology are transforming healthcare practices inside and outside hospitals, benefiting both medical staff and patients (Kim et al. 2018).
- Smart Fibers and Textiles for Personal Thermal Management: Advancements in smart fibers and textiles for personal thermal man-

agement are enhancing comfort and efficiency in healthcare applications (Zuo et al. 2023).

- Wearable Sensors for Remote Monitoring in Telemedicine Healthcare: Wearable sensors integrated with flexible electronics and medical IoT technologies are enabling remote health monitoring and telemedicine services (Kalasin and Surareungchai 2023).
- Blockchain for IoT Applications in Smart Healthcare: Blockchain technology is being utilized in smart healthcare applications to enhance security, interoperability, and scalability in healthcare systems (Abdelmaboud et al. 2022).
- Deep Learning-based Fall Detection for IoT-enabled Smart Healthcare: Cloud computing, AI, and IoT technologies are transforming traditional healthcare systems into smart healthcare, improving patient care and outcomes for aging patients (Basahel et al. 2023).

Smart healthcare leverages technology-enabled innovations to transform healthcare delivery, enhance patient outcomes, and optimize resource utilization across the healthcare domain. Utilizing the smart healthcare solutions can lead to improved efficiency, accessibility, and quality of care while addressing evolving patient needs and healthcare challenges in the digital age.

5.8 Healthcare 5.0

The concept of Healthcare 5.0 involves the integration of advanced technologies, data analytics, and innovative approaches to revolutionize healthcare services. Healthcare 5.0 represents the next evolution in healthcare delivery, characterized by the integration of advanced technologies, data analytics, and innovative approaches to enhance patient care, improve outcomes, and optimize healthcare services. This concept emphasizes a holistic and patient-centered approach to healthcare, focusing on personalized, preventive, and proactive strategies to address individual health needs. Healthcare 5.0 aims to leverage cutting-edge technologies such as AI, IoT, robotics, and wearable devices to transform healthcare delivery, improve efficiency, and enhance the overall quality of care provided to patients. Various aspects of Healthcare 5.0 are identified and defined below with specific details in the subsections.

- NLP in Healthcare AI: NLP has been significantly utilized to extract crucial information from unstructured data to enhance decision-making, administrative reporting, and research in healthcare AI (Wen et al. 2019).
- IoT and Robotics in Healthcare: The utilization of IoT and robotics in healthcare services, such as rehabilitation, assistive surgery, elderly care, and prosthetics, demonstrates the transformative potential of these technologies in healthcare delivery (Pradhan et al. 2021).
- Physical Environment in Healthcare Facilities: The influence of the physical environment on the health and well-being of patients and healthcare workers underscores the importance of healthcare facility design in fostering positive health outcomes (Chi et al. 2020).
- Plasma Medicine and Nanotechnology: The integration of plasma, nano, and digital technologies in healthcare is known to enhance patient outcomes and comfort, and alleviate healthcare system burdens, highlighting the promise of multimodal healthcare packages (Liu et al. 2020).
- Wearable Healthcare Devices: The design and advancement of wearable healthcare devices for monitoring physical, electrophysiological, chemical, and biological signals showcase the progress in wearable technology for healthcare monitoring (Lyu et al. 2021).

5.8.1 Remote Patient Monitoring

AI-based techniques are being utilized to monitor patients remotely, and the following is a list of examples of the utilization of AI for remote patient monitoring.

1. Patients' Perception of Wearable Devices and AI in Healthcare: Wearable biometric monitoring devices enable remote, high-frequency monitoring of patients' health outside the hospital, offering insights into patients' views on wearable devices and AI in healthcare (Tran et al. 2019).
2. Improvements in Patient Monitoring in Intensive Care Units: Remote patient monitoring allows clinicians to collect health data via vital sign sensors and provide recommendations for managing patients, enhancing patient monitoring in intensive care settings (Poncette et al. 2020).

3. Utility of Remote Monitoring in Implantable Cardiac Devices: Remote monitoring of implantable cardiac devices reduces the time frame for clinical decisions, leading to improved patient outcomes (Boriani et al. 2023; Alotaibi et al. 2020; Bekfani et al. 2021).

5.8.2 Assisted Surgery and Surgical Robots

AI-based assisted surgery and surgical robots are transforming the field of medicine by integrating advanced technologies to enhance surgical procedures, improve precision, and optimize patient outcomes. Following are some of the examples of AI-based assisted surgery and surgical robots:

1. Surgical robots are increasingly utilized in various surgical specialties such as gynecology, urology, orthopedics, and general surgery, offering enhanced capabilities and precision in surgical procedures (Zhang et al. 2021).
2. Robotic-assisted surgery systems, such as the da Vinci robot, rely on direct vision through laparoscopic cameras for surgical guidance, enhancing visualization and precision in surgical procedures (Kwartowitz et al. 2006).
3. Surgical robots have the potential to enhance surgical accuracy by removing physiological tremors, improving manual dexterity, and providing increased precision in microsurgical procedures (Tan et al. 2018).
4. Robotic-assisted surgery is transforming procedures in head and neck surgery, offering improved precision, minimally invasive techniques, and enhanced outcomes for patients (Garas and Arora 2018).
5. The design of surgical robots focuses on conducting or assisting in surgical procedures, providing surgeons with advanced tools and technologies to improve surgical outcomes (Swarnalatha and Menon 2022).
6. Robotic-assisted laparoscopic surgery, such as robotic radical prostatectomy, has become a primary approach in the surgical management of localized prostate cancer, leading to improved patient care (Darlington et al. 2022).

7. The evolution of surgical robotics has led to advancements in surgical techniques, improved patient outcomes, and enhanced collaboration between surgeons and robotic systems (Najarian and Afshari 2012).
8. The combination of AI and robotics is transforming surgical procedures, enabling robots to learn and perform routine operative tasks under human supervision, leading to improved surgical outcomes (O'-Sullivan et al. 2019).
9. Modern thoracic surgery and other surgical specialties are increasingly utilizing robotic surgical systems, which offer improved safety, efficiency, and precision in minimally invasive procedures (Najafi et al. 2023).
10. AI-enabled surgical robots are assisting surgeons in performing complex procedures with greater control and precision, showcasing the potential of technology in enhancing surgical capabilities (Shaheen 2021).
11. AI-driven technologies are being integrated into surgical robots for tasks such as bone-drilling state recognition, hip fracture fixation, and dynamic motion scaling, enhancing surgical accuracy and patient outcomes (Xia et al. 2022).
12. Scalable, learning-based X-ray image analysis systems are being developed to enhance surgical tool detection, hip image analysis, and COVID-19 lung lesion segmentation, showcasing the potential of AI in surgical robotics (Gao et al. 2022).

Recent developments in AI systems and ML models have paved the way for the next generation of surgical robots that can autonomously learn and perform various tasks with human oversight, enhancing surgical precision and efficiency (Andras et al. 2020). The combination of robots and AI is expected to elevate surgical procedures to the next level, providing advanced capabilities and improved outcomes for patients (Kawashima et al. 2019). As robotic surgeries advance toward autonomy, ethical discussions are essential to address decision-making, infrastructure requirements, and legal obligations in the field of surgical robotics (Chang et al. 2021). Also, cybersecurity considerations, procedural care protection, and the impact of robotic malfunctions are important factors to address in the implementation of robotic surgery systems (Gordon et al. 2022). The future of surgical robotics involves the incorporation of AI-driven technolo-

gies to enable autonomous actions in surgery, potentially reducing operation time, increasing accuracy, optimizing surgical procedures, developing cognitive engines, improving automation, and human–robot collaboration to enhance surgical procedures and patient care (Chui et al. 2014; Feizi et al. 2021).

5.8.3 Drug Design and Development

AI is transforming drug design and development by utilizing advanced technologies to accelerate the discovery of new therapeutic compounds, optimize treatment strategies, and improve patient care. Following are some of the applications of AI in drug design and development:

1. Computational biology, computer-aided drug design, and AI are enhancing the efficiency of drug discovery by reducing time and costs, providing a promising approach to streamline the drug development process (Zhang et al. 2022).
2. AI is pivotal in various processes such as drug discovery, disease identification, personalized treatment, clinical trial research, and pharmacological studies, demonstrating its potential in advancing pharmacology research and practice (Kumar et al. 2023). The incorporation of AI in pharmacology enables model-informed precision medicine, facilitating precise, efficient, and inclusive drug development and utilization through advanced data analytics and AI technologies (Terranova and Venkatakrishnan 2024).
3. AI models, such as unsupervised clustering and supervised ML, are employed in drug discovery to identify potential drug compounds and suitable patient populations and enhance therapeutic drug monitoring, thereby improving treatment outcomes (van der Lee and Swen 2023).
4. AI tools are repurposed for disease modeling, for instance, in identifying face recognition deficits in neurodegenerative diseases, to personalize adaptive dosing protocols and enhance treatment outcomes, showcasing the potential of AI in pharmacology studies (Singh and Ramanathan 2023).
5. The development of network-based AI methods in network pharmacology is crucial for elucidating treatment mechanisms of complex dis-

eases, utilizing AI technology to analyze extensive genomics data and advance traditional medicine research (Zhang et al. 2024).
6. The practical application of AI-based deep learning and cloud computing in traditional and Western medicine is reshaping the diagnosis and treatment of rheumatoid arthritis, optimizing disease management and treatment models (Wang et al. 2021).

The aforementioned examples and advancements in AI-driven drug design and development highlight the significant impact of AI in revolutionizing pharmacology research, precision medicine, and therapeutic outcomes.

5.8.4 Assisted Living

AI is playing a significant role in assisted living environments, offering innovative solutions to support independent living, enhance quality of life, and improve overall well-being. Following are examples of the usage of AI for assisted living:

1. Technologies such as the IoT, ambient/active assisted living (AAL) robots, and AI have shown great potential in promoting independent living, improving mental and physical health, and enhancing the quality of life for older adults (Wang et al. 2019).
2. Patient and public involvement in the transition to AI-assisted healthcare is crucial for designing justice-based AI for diverse patient needs, ensuring inclusivity and ethical considerations in AI applications (Zidaru et al. 2021).
3. AI-based AAL technologies, incorporating sensor networks, monitoring systems, telehealth technologies, wearable sensors, and assistive robotics, aim to support elderly individuals in their daily activities and extend their ability to live independently, enhancing their quality of life and well-being (Hartmann et al. 2023; Holzinger et al. 2022).

The integration of robots in assisted living environments requires a multidisciplinary approach to address various research areas and create comprehensive solutions to support individuals in assisted living settings (Mahmood et al. 2021). AI is revolutionizing assisted living environments,

promoting independence, enhancing care services, and improving the overall quality of life for individuals in need of support.

5.8.5 Cancer Diagnostics and Treatment

AI is transforming cancer diagnostics and treatment by utilizing advanced technologies to improve early detection, treatment optimization, and patient outcomes. Following are examples of the usage of AI in cancer diagnostics and treatment:

1. AI algorithms and ML models are employed to analyze intricate patterns in medical data, enabling early and precise cancer diagnosis, resulting in timely interventions and enhanced patient survival rates (Govindan et al. 2023).
2. AI technologies are facilitating precision medicine in cancer treatment by enhancing drug delivery to specific tumor sites, reducing side effects, and customizing treatment regimens based on individual patient characteristics (Govindan et al. 2023).
3. AI has the potential to transform cancer research by addressing limitations in traditional diagnostics and treatment, opening doors to innovative approaches in cancer healthcare (Sebastian and Peter 2022).
4. In the field of urology, AI is being trialed and implemented to assist in the diagnosis and treatment of prostate cancer, providing personalized and efficient treatment strategies for patients (Chu et al. 2023, Ström et al. 2020).
5. AI-powered targeted drug delivery systems are enhancing cancer therapy outcomes by optimizing drug administration to specific tumor sites, reducing systemic side effects, and improving treatment effectiveness (Das 2023, Kumar and Saha 2022).

AI technologies show potential in advancing cancer diagnostics, treatment decision-making, and patient outcomes, with the capacity to significantly enhance the quality of cancer care in the future (Ali et al. 2023).

5.9 Ethics, Bias, and Fairness Constraints

As discussed, AI in healthcare holds great promise for improving patient outcomes, enhancing diagnostic accuracy, and optimizing treatment plans. However, ethical considerations, bias, and fairness constraints are critical issues that must be addressed to ensure the responsible and equitable use of AI techniques in healthcare. Following are some of the ethics, bias, and fairness constraints that are essential to address for the effective utilization of AI in healthcare:

- Patient Apprehensions about AI in Healthcare: Proactively addressing concerns is crucial for ethical innovation and ensuring the long-term success of AI applications in healthcare (Richardson et al. 2021).
- Ethical Implications of AI in Healthcare: Common ethical concerns include privacy, trust, accountability, responsibility, and bias, which need to be addressed in the adoption of AI in healthcare (Murphy et al. 2021; Pasricha 2022). Furthermore, security, discrimination, transparency, informed consent, and human interaction are key ethical implications of AI use in healthcare that need to be managed (Amedior 2023).
- Ethical Considerations of AI and Robotics in Healthcare: Addressing issues such as privacy, data security, bias, fairness, accountability, transparency, autonomy, and human oversight is essential for the ethical implementation of AI and robotics in healthcare (Elendu et al. 2023).
- Algorithmic Bias in Healthcare AI: Mitigating algorithmic bias in healthcare AI is crucial to prevent discrimination and inequity, especially when AI systems are trained on nonrepresentative data (Redrup Hill et al. 2023). Stakeholders have endorsed various strategies to mitigate algorithmic bias in healthcare AI, raising ethical questions about responsibility for bias and its impact (Aquino et al. 2023). Furthermore, AI models that are trained on nonrepresentative datasets may not have the ability to generalize well to diverse populations, leading to biased outcomes.
- Embedded Ethics in Medical AI Development: An “embedded ethics” approach involving ethicists and developers can effectively address ethical issues in the practical development of medical AI, ensuring eth-

ical considerations are integrated from the outset (McLennan et al. 2022).

- Fairness and Bias in AI: Addressing bias in AI requires diverse datasets, transparency, accountability, and exploring alternative AI paradigms that prioritize fairness and ethical considerations (Ferrara 2023).
- Measurement and Mitigation of Bias in AI: Understanding and mitigating bias in AI systems is crucial for ensuring fairness, transparency, and ethical use of AI in healthcare (Gray et al. 2024). Historical data that reflects existing inequalities and discrimination can lead to biases in the AI model if these existing biases are not addressed properly.

Ensuring ethics, bias, and fairness in AI for healthcare requires a multifaceted approach involving diverse stakeholders, robust regulatory frameworks, continuous monitoring, and adaptive technologies.

5.9.1 Addressing Ethics, Bias, and Fairness Constraints

Some of the ways by which these ethics, bias, and fairness constraints can be addressed are:

1. Implementing fairness metrics and continuous monitoring of the AI models can help identify and mitigate the biases early in the AI modeling process. Fairness-aware ML techniques can be leveraged to ensure balanced performance across diverse groups (Kheya et al. 2024; Caton and Haas 2024).
2. Utilizing and building diverse groups of developers, clinicians, and ethicists can help identify and address ethical issues early in the design and development phases (McLennan et al. 2022; Tang et al. 2023a).
3. Engaging various stakeholders, such as patients, and communities, can ensure that the AI tools meet the needs of a diverse group of individuals and address ethical concerns (Morley et al. 2020).
4. Regular audits and evaluations of the AI systems should be performed to detect biases, assess fairness, and ensure ethical compliance (Pessach and Shmueli 2022).
5. Adaptive algorithms with the ability to learn and correct biases over time using continuous feedback and data updates need to be devel-

oped (Kheya et al. 2024).

6. Establishing clear regulatory guidelines on data usage, privacy, bias mitigation, and accountability is critical for the use of AI in healthcare (Chen et al. 2023).
7. Policymakers need to work with technologists and healthcare professionals to develop policies that promote the development and use of ethical AI (World Health Organization 2021).

5.10 Concluding Remarks

The integration of AI in healthcare represents a monumental shift toward more precise, efficient, and personalized medical care. AI's ability to analyze vast amounts of data quickly and accurately offers unprecedented opportunities for early diagnosis, targeted treatments, and improved patient outcomes. From AI-powered imaging systems that detect diseases at their earliest stages to advanced diagnostic tools that provide real-time insights, the applications of AI are revolutionizing every aspect of healthcare. Moreover, AI enhances the capabilities of healthcare professionals, allowing for more accurate interpretations of medical data, streamlined workflows, and reduced human error. The benefits extend to patients, who receive faster diagnoses, more effective treatments, and continuous monitoring that can preempt complications.

While AI holds tremendous potential to revolutionize healthcare by improving diagnostics, personalizing treatment, enhancing operational efficiency, and accelerating drug discovery, it also brings significant challenges. Addressing issues related to data privacy, algorithmic bias, regulatory frameworks, and ethical considerations is crucial for the responsible and equitable implementation of AI in healthcare. As the technology continues to evolve, ongoing collaboration among healthcare providers, technologists, regulators, and ethicists will be essential to harness the full benefits of AI while mitigating its risks. Continuous education and training for healthcare professionals are essential to effectively integrate AI into clinical practice.

5.11 Future Directions

The future of healthcare is undeniably intertwined with AI, paving the way for innovations that could significantly enhance the quality of life and healthcare delivery worldwide. AI will continue to play a pivotal role in advancing precision medicine, enabling personalized treatment plans based on individual genetic, environmental, and lifestyle factors (Rajpurkar et al. 2022). Moreover, overcoming challenges such as the translation of AI systems from research to clinical practice, intrinsic ML limitations, and sociocultural barriers will be crucial for the widespread adoption of AI in healthcare (Kelly et al. 2019; Vaishya et al. 2020). AI-driven precision medicine technologies will continue to transform patient care by providing more accurate and effective strategies for diagnosing, preventing, and treating diseases (Charow et al. 2021).

The adoption of AI in public healthcare is steadily increasing, driven by potential cost savings and the efficiency of AI technologies in replacing face-to-face interactions in healthcare service delivery (Sun and Medaglia 2019). AI technologies will transform the practice of medicine by democratizing and standardizing connected and AI-augmented care, precision diagnostics, therapeutics, and ultimately precision medicine (Vaishya et al. 2020). Furthermore, designing AI-augmented healthcare delivery systems that prioritize physician buy-in and patient acceptance will be crucial for successfully adopting and integrating AI technologies in healthcare settings (Dai and Tayur 2022). As AI adoption in healthcare grows, empowering healthcare professionals to develop, validate, and implement AI-enabled technologies will be essential for the successful integration and utilization of AI in healthcare (Esmaeilzadeh 2020). AI adoption will be able to uplift the overall care process by standardizing clinical decisions and procedures, leading to enhanced and safe healthcare delivery (Cheng et al. 2022). Moreover, parental attitudes toward AI-driven precision medicine technologies in pediatric healthcare will shape the adoption and acceptance of AI applications in pediatric medicine (Sisk et al. 2020).

Future adoption of AI in healthcare will require addressing ethical and legal implications to ensure patient privacy, data security, and compliance

with regulations (Brown et al. 2010). Building trust in AI-based systems and devices by ensuring reliability, integrity, and accuracy will be critical for fostering patient trust and acceptance of AI technologies in healthcare (Tursunbayeva and Renkema 2023), and establishing trust in AI-based systems is paramount for their successful implementation.

These future directions for AI in healthcare are supported by recent research highlighting the potential of AI to revolutionize healthcare practices and outcomes by improving patient care, enhancing healthcare delivery, and advancing precision medicine (Bajwa et al. 2021; Chikhaoui et al. 2022; Sisk et al. 2020).

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